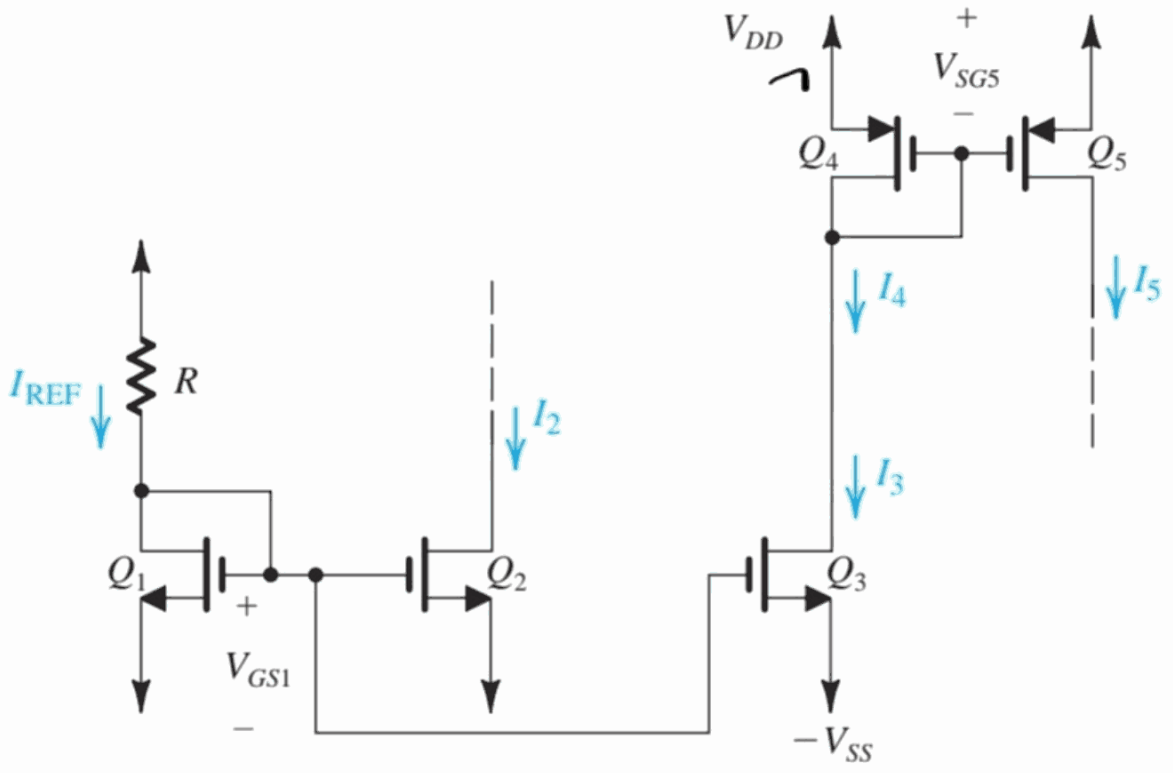
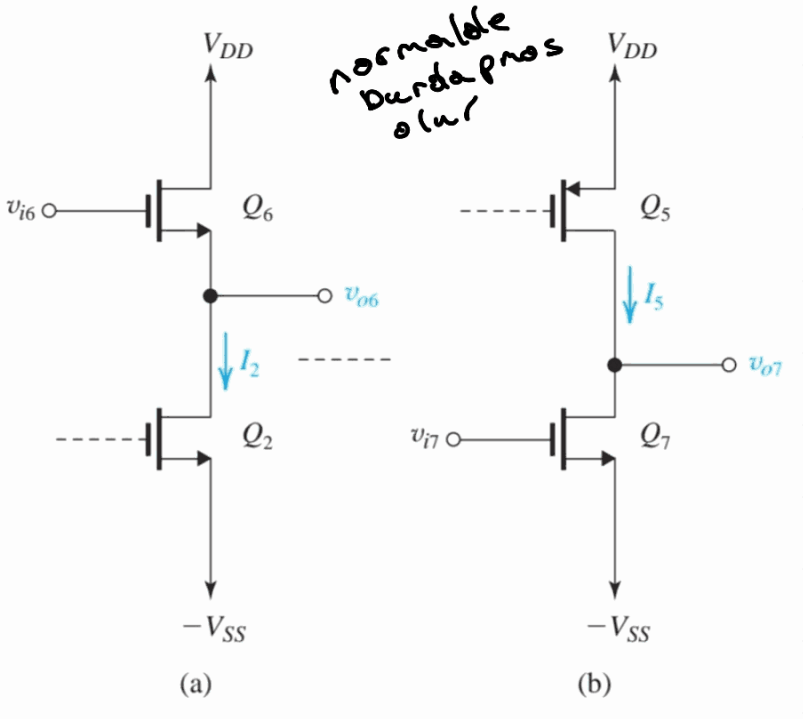
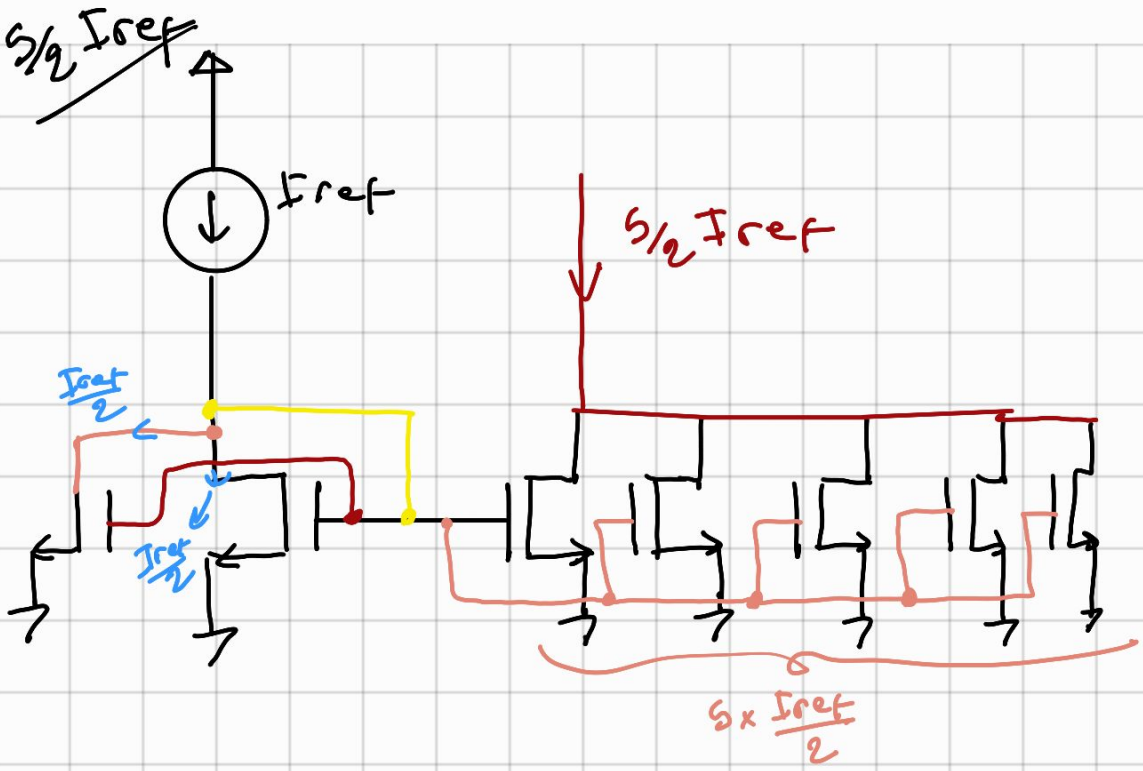
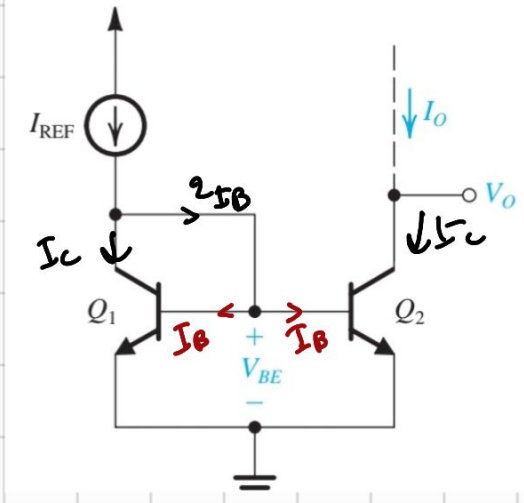

V_{DD}
+
V_{SG5}
−
Q_4
Q_5
I_{REF}
R
I_2
I_4
I_5
I_3
Q_1
Q_2
Q_3
+
V_{GS1}
−
$-V_{SS}$


V_{DD}
normalde Durdapnos olur
v_{i6}
Q_6
v_{o6}
I_2
Q_2
$-V_{SS}$
(a)
V_{DD}
Q_5
I_5
v_{o7}
v_{i7}
Q_7
$-V_{SS}$
(b)



BJT Circuits



$I_C = \beta I_B$

$I_B = I_C/\beta$

$I_{ref} = I_C + 2I_B$

$I_{ref} = I_C\left(1 + \frac{2}{\beta}\right)$

$I_O = I_C$

$$\boxed{\frac{I_O}{I_{ref}} = \frac{1}{(1 + 2/\beta)}} \simeq 1$$

β büyünce akımaynasın kalitesi artar

what happens if

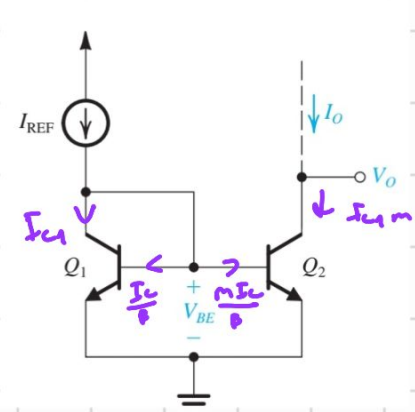
EBA = 1 EBA2 = m

$I_{C_1} = I_S e^{V_{BE}/V_T}$

$I_{C_2} = mI_S e^{V_{BE}/V_T}$

$I_{C_1} m = I_{C_2}$

⇒



$I_{ref} = I_{C1} + \frac{(1+m)I_{C1}}{\beta}$

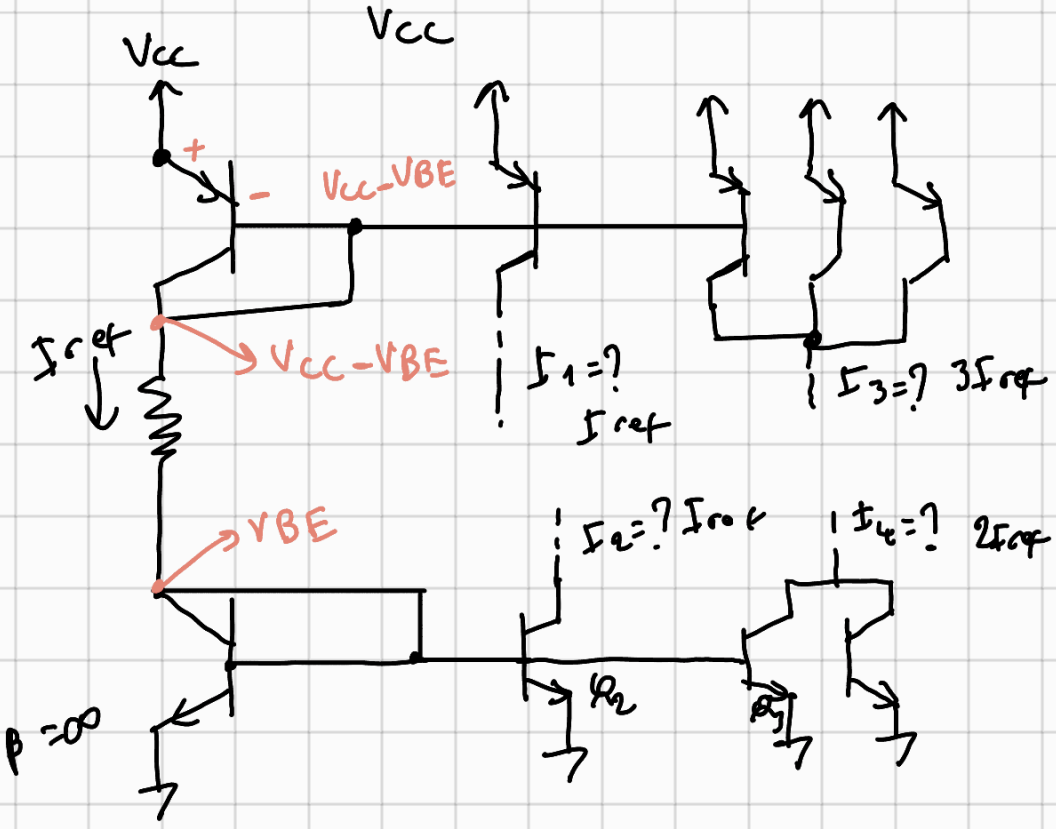
$= I_{C1}\left(1 + \frac{(m+1)}{\beta}\right)$

$I_O = I_{C1}m$

$\frac{I_O}{I_{ref}} = \frac{m}{1 + \frac{(m+1)}{\beta}}$

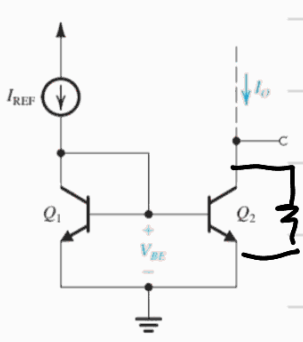
$$\frac{I_O}{I_{ref}} = \frac{m}{1 + \frac{(m+1)}{\beta}}$$

Current Steering BJT



$$I_{ref} = \frac{(V_{CC} - V_{BE}) - V_{BE}}{R}$$

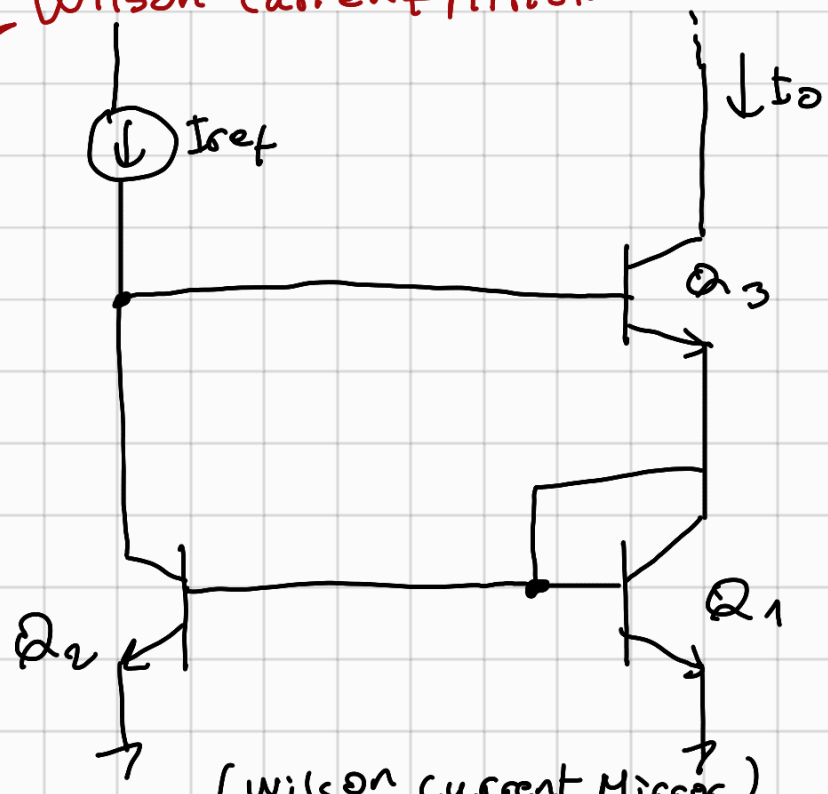
$$= \frac{V_{CC} - 2V_{BE}}{R} = \frac{V_{CC} - V_{BE} - V_{EB}}{R}$$



$$r_o = \frac{V_A}{I_C} \Rightarrow \text{BJT}$$

$$r_o = \frac{V_A}{I_D} = \frac{V_A' L}{I_D} \Rightarrow \text{Mosfet}$$

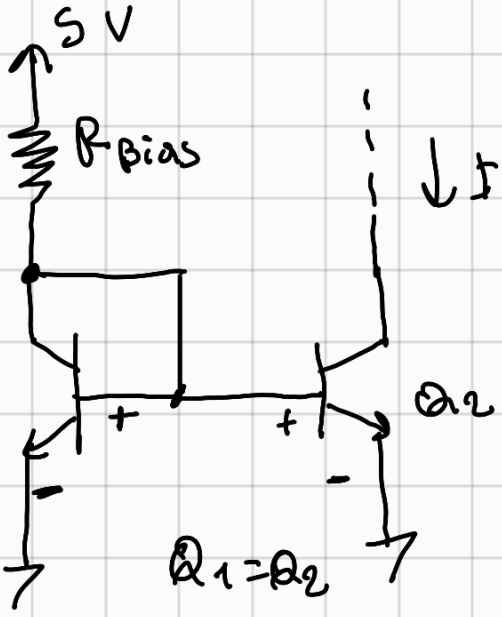
★ Wilson Current Mirror



(Wilson current Mirror)

$\frac{I_o}{I_{ref}} = ?$

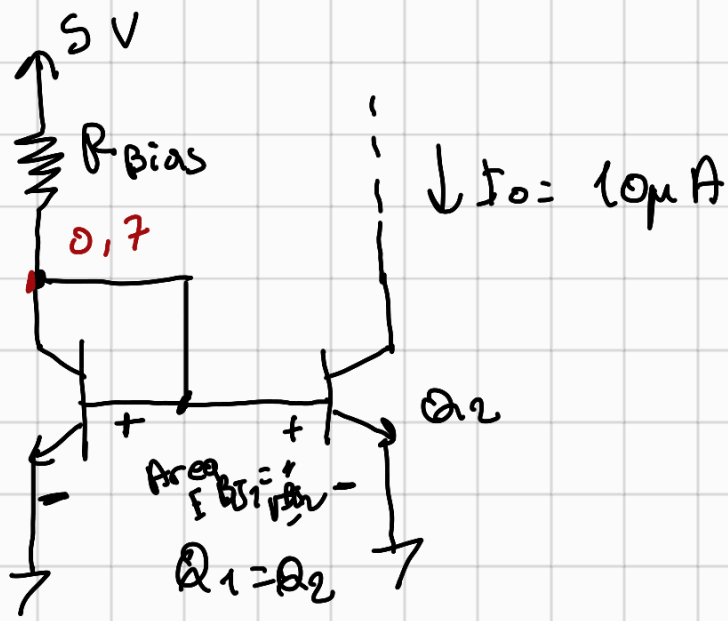
Bob Widlar Current Source



What is the value of R_{Bias}

$\beta = 100$

$\beta = \infty$

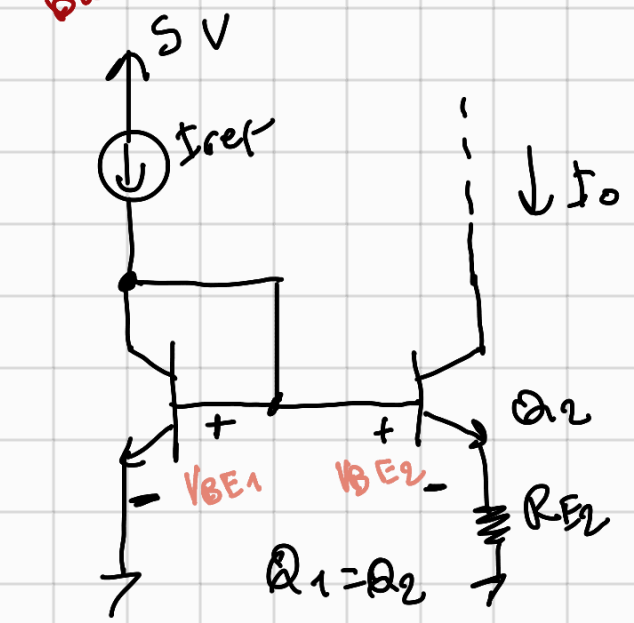


$R_{BIAS} = \frac{4,3}{10\mu A}$ $\quad R_{BIAS} \Rightarrow 430k\Omega$ $(\beta = \infty)$

$R_{BIAS} = 421,6k\Omega$ $(\beta = 100)$

$$\frac{I_o}{I_{ref}} = \frac{1}{1+2/\beta}$$

Bob widlar bunu önermiş



if $\beta = \infty$

$$I_{ref} = I_s e^{\frac{V_{BE1}}{V_T}}$$

$$I_o = I_s e^{\frac{V_{BE2}}{V_T}}$$

$$V_{BE1} = V_{BE2} + I_o \cdot R_{E2}$$

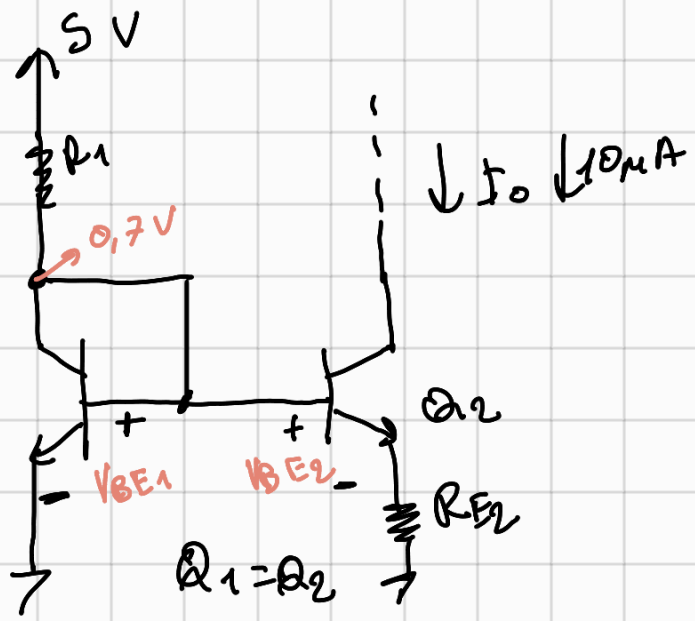
$$V_{BE1} - V_{BE2} = I_o R_{E2}$$

$$\frac{I_o}{I_{ref}} = \frac{e^{\frac{V_{BE2}}{V_T}}}{e^{\frac{V_{BE1}}{V_T}}}$$

$$\ln\left(\frac{I_o}{I_{ref}}\right) = \frac{V_{BE2} - V_{BE1}}{V_T}$$

$$V_{BE2} - V_{BE1} = V_T \ln\left(I_o / I_{ref}\right) = -I_o \cdot R_{E2}$$

$$I_o = \frac{V_T \ln\left(I_{ref}/I_o\right)}{R_{E2}}$$



$R_1 = ?$

$R_2 = ?$

ΔR_{total} with previous design

$$\frac{5 - 0,7}{R_1} = I_{ref} = \frac{4,3}{R_1}$$

$$I_o = \frac{V_T \cdot \ln\left(\frac{4,3}{I_o R_1}\right)}{R_{E2}}$$

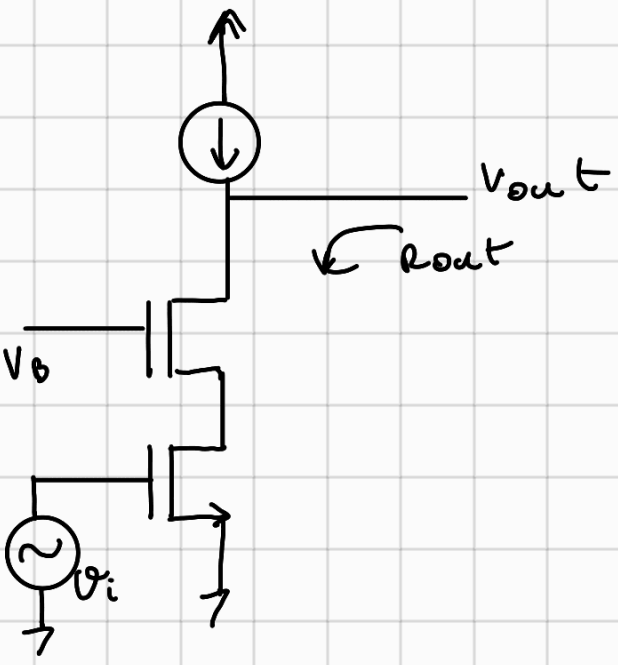
$$10\mu A = \frac{25\,mV \cdot \ln\left(\frac{I_{ref}}{10\mu A}\right)}{R_2}$$

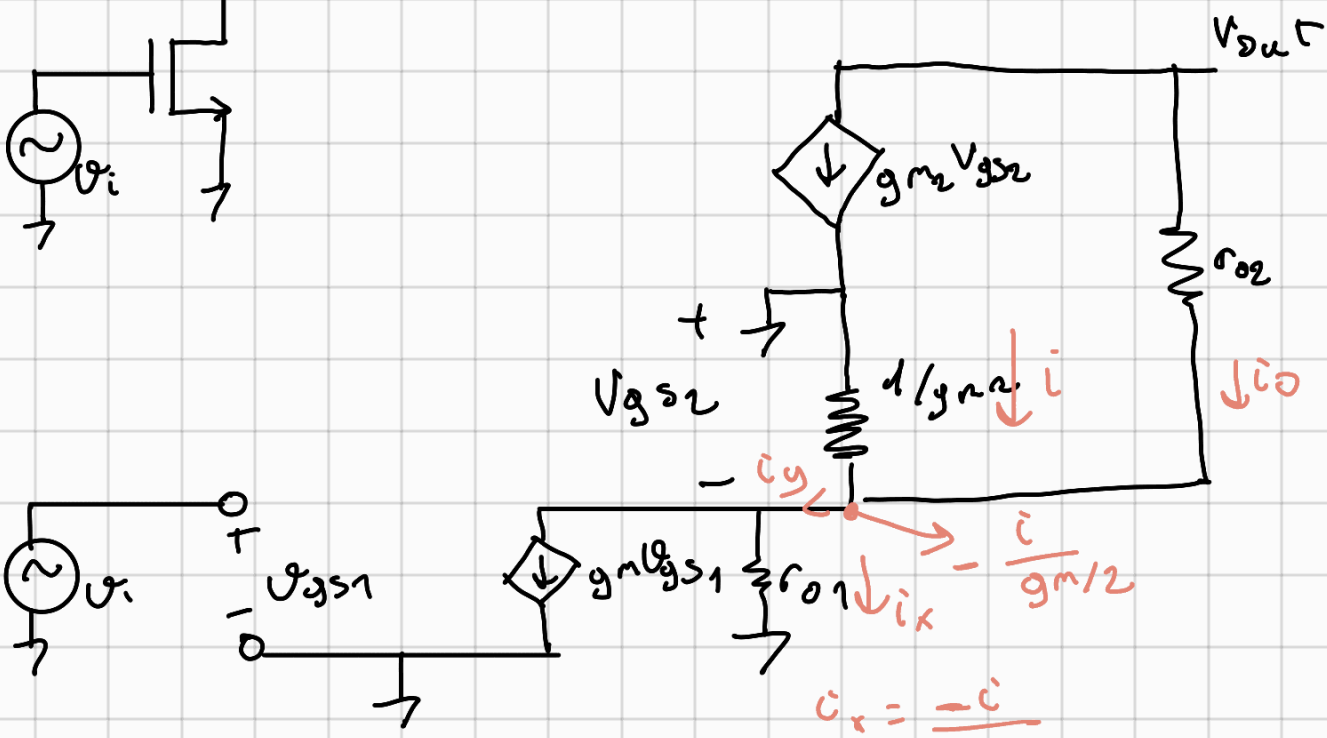
$$I_{ref} = \frac{5 - 0,7}{R_1} \rightarrow 100K$$

$$I_{ref} = 43\,mA$$

$$R_2 = \frac{\ln(4{,}3)\ 25\text{mV}}{10^{-5}\ 10^{-2}} \rightarrow R_2 = 3{,}6\,k\Omega$$

7.3 The Cascode Amplifier





$$i_x = \frac{-i}{r_{o1} g_{m2}}$$

$$v_i = V_{gs1}$$

$$i_y = \frac{-i}{r_{o1} g_{m2}} + g_m v_i$$

$$i_o = \frac{-i}{r_{o1} g_{m2}} + g_{m1} v_i - i$$

$$i_o = \frac{V_{out} - r_{o1}\, i_x}{r_{o2}} = \frac{V_{out} - V_{gs2}}{r_{o2}}$$

$$v_{out} = r_{o2}\, i_o + \frac{-i}{g_{m2}}$$

$$v_{out} = r_{o2}\left[\frac{-i}{r_{o1} g_{m2}} + g_{m1} v_i - i\right] - \frac{i}{g_{m2}}$$

$g_{m_1} = g_{m_2}$

$$\frac{v_{out}}{v_i} = \frac{r_o \left[\frac{-i}{g_m r_{o_1}} + g_m v_i \right] - \frac{i}{g_m}}{v_i}$$

Ödev

$$\rightarrow \quad i_o = \frac{-g_{m_2} v_{gs_2}}{r_o g_{m_2}} + g_{m_1} v_i - g_{m_2} v_{gs_2}$$

$$g_{m_1} v_i' = i_o + \frac{v_{gs_2}}{r_{o_2}} + g_{m_2} v_{gs_2}$$

$$R_o \simeq (g_{m_2} r_{o_2}) r_{o_1}$$

$$A_{v_o} = \frac{v_o}{v_i} = -g_{m_1} R_o$$

$$A v_o = -(g_{m1} r_{o1})(g_{m2} r_{o2})$$

For the case $g_{m1} = g_{m2} = g_m$ and $r_{o1} = r_{o2} = r_o$

$$A_v = -$$

$$($$